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SYSTEMS AND METHODS FOR CONSOLIDATING SYMBOLS IN ELECTRONIC GAMING

Abstract

An electronic gaming device including a display device, a memory device storing instructions, and a processor in communication with the display device and the memory device is described. The instructions, when executed by the processor, cause the processor to cause display of a plurality of symbols, present an output amount, determine that the plurality of symbols includes a plurality of consolidating symbols, cause display of the plurality of consolidating symbols consolidating into a consolidated symbol, fill any vacant symbol positions with new symbols, determine a second output amount, and present the second output amount. The instructions also cause the processor to cause display of a third plurality of symbols, the third plurality of symbols and the consolidated symbol defining a fourth plurality of symbols, determine a third output amount based upon the fourth plurality of symbols, and present the third output amount.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of and claims priority to U.S. patent application Ser. No. 17/956,339, filed Sep. 29, 2022, which is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The field of disclosure relates generally to electronic gaming, and more specifically, to systems and methods for consolidating symbols in electronic gaming.

BACKGROUND

[0003] Electronic gaming machines (“EGMs”) or gaming devices provide a variety of wagering games such as slot games, video poker games, video blackjack games, roulette games, video bingo games, keno games and other types of games that are frequently offered at casinos and other locations. Play on EGMs typically involves a player establishing a credit balance by inputting money, or another form of monetary credit, and placing a monetary wager (from the credit balance) on one or more outcomes of an instance (or single play) of a primary or base game. In some cases, a player may qualify for a special mode of the base game, a secondary game, or a bonus round of the base game by attaining a certain winning combination or triggering event in, or related to, the base game, or after the player is randomly awarded the special mode, secondary game, or bonus round. In the special mode, secondary game, or bonus round, the player is given an opportunity to win extra game credits, game tokens or other forms of payout. In the case of “game credits” that are awarded during play, the game credits are typically added to a credit meter total on the EGM and can be provided to the player upon completion of a gaming session or when the player wants to “cash out.”

[0004] “Slot” type games are often displayed to the player in the form of various symbols arrayed in a row-by-column grid or matrix. Specific matching combinations of symbols along predetermined paths (or paylines) through the matrix indicate the outcome of the game. The display typically highlights winning combinations/outcomes for identification by the player. Matching combinations and their corresponding awards are usually shown in a “pay-table” which is available to the player for reference. Often, the player may vary his/her wager to include differing numbers of paylines and/or the amount bet on each line. By varying the wager, the player may sometimes alter the frequency or number of winning combinations, frequency or number of secondary games, and/or the amount awarded.

[0005] Typical games use a random number generator (RNG) to randomly determine the outcome of each game. The game is designed to return a certain percentage of the amount wagered back to the player over the course of many plays or instances of the game, which is generally referred to as return to player (RTP). The RTP and randomness of the RNG ensure the fairness of the games and are highly regulated. Upon initiation of play, the RNG randomly determines a game outcome and symbols are then selected which correspond to that outcome. Notably, some games may include an element of skill on the part of the player and are therefore not entirely random.

BRIEF DESCRIPTION

[0006] In one aspect, an electronic gaming device is described. The electronic gaming device includes a display device, a memory device storing instructions, and a processor in communication with the display device and the memory device. The instructions, when executed by the processor, cause the processor to cause display of a plurality of symbols at a plurality of symbol positions on

the display device wherein the plurality of symbols correspond to an output amount for a play of an electronic game, present the output amount for the play of the electronic game, and determine that the plurality of symbols includes a plurality of consolidating symbols. The instructions also cause the processor to cause display of the plurality of consolidating symbols consolidating into a consolidated symbol at a symbol position of the plurality of symbol positions previously occupied by a consolidating symbol wherein the consolidated symbol is associated with a number of future plays of the electronic game, fill any vacant symbol positions of the plurality of symbol positions with new symbols wherein the new symbols, the consolidated symbol, and any remaining symbols of the plurality of symbols define a second plurality of symbols, and determine a second output amount for the play of the electronic game based upon the second plurality of symbols. The instructions further cause the processor to present the second output amount for the play of the electronic game, for each symbol position of the plurality of symbol positions other than the symbol position including the consolidated symbol, cause display of a third plurality of symbols for one of the number of future plays, the third plurality of symbols and the consolidated symbol defining a fourth plurality of symbols, determine a third output amount based upon the fourth plurality of symbols, and present the third output amount for the one of the number of future plays.

[0007] In another aspect, a non-transitory computer-readable storage medium with instructions stored thereon is described. The instructions, in response to execution by a processor, cause the processor to cause display of a plurality of symbols at a plurality of symbol positions on a display device wherein the plurality of symbols correspond to an output amount for a play of an electronic game, transmit the output amount for the play of the electronic game, and determine that the plurality of symbols includes a plurality of consolidating symbols. The instructions also cause the processor to cause display of the plurality of consolidating symbols consolidating into a consolidated symbol at a symbol position of the plurality of symbol positions previously occupied by a consolidating symbol wherein the consolidated symbol is associated with a number of future plays of the electronic game, cause any vacant symbol positions of the plurality of symbol positions to be filled with new symbols wherein the new symbols, the consolidated symbol, and any remaining symbols of the plurality of symbols define a second plurality of symbols, determine a second output amount for the play of the electronic game based upon the second plurality of symbols, and transmit the second output amount for the play of the electronic game. The instructions further cause the processor to, for each symbol position of the plurality of symbol positions other than the symbol position including the consolidated symbol, cause display of a third plurality of symbols for one of the number of future plays, the third plurality of symbols and the consolidated symbol defining a fourth plurality of symbols, determine a third output amount based upon the fourth plurality of symbols, and transmit the third output amount for the one of the number of future plays.

[0008] In yet another aspect, a method of electronic gaming implemented by a processor in communication with a memory is described. The method includes causing display of a plurality of symbols at a plurality of symbol positions on a display device wherein the plurality of symbols correspond to an output amount for a play of an electronic game, presenting the output amount for the play of the electronic game, and determining that the plurality of symbols includes a plurality of consolidating symbols. The method also includes causing display of the plurality of consolidating symbols consolidating into a consolidated symbol at a symbol position of the plurality of symbol positions previously occupied by a consolidating symbol, wherein the consolidated symbol is associated with a number of future plays of the electronic game, causing any vacant symbol positions of the plurality of symbol positions to be filled with new symbols wherein the new symbols, the consolidated symbol, and any remaining symbols of the plurality of symbols define a second plurality of symbols, determining a second output amount for the play of the electronic game based upon the second plurality of symbols, and presenting the second output amount for the play of the electronic game. The method further includes, for each symbol position of the plurality

of symbol positions other than the symbol position including the consolidated symbol, causing display of a third plurality of symbols for one of the number of future plays, the third plurality of symbols and the consolidated symbol defining a fourth plurality of symbols, determining a third output amount based upon the fourth plurality of symbols, and presenting the third output amount for the one of the number of future plays.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is an exemplary diagram showing several EGMs networked with various gaming related servers.

[0010] FIG. 2A is a block diagram showing various functional elements of an exemplary EGM.

[0011] FIG. 2B depicts a casino gaming environment according to one example.

[0012] FIG. 2C is a diagram that shows examples of components of a system for providing online gaming according to some aspects of the present disclosure.

[0013] FIG. 3 illustrates, in block diagram form, an implementation of a game processing architecture algorithm that implements a game processing pipeline for the play of a game in accordance with various implementations described herein.

[0014] FIG. 4 illustrates an example screenshot and/or interface of an electronic game before symbols are consolidated, in accordance with the present disclosure and for use with the architectures, systems, and/or environments shown in FIGS. 1-3.

[0015] FIG. 5 illustrates an example screenshot and/or interface of the electronic game shown in FIG. 4 after symbols are consolidated.

[0016] FIG. 6 illustrates a first example screenshot and/or interface of a sequence of the electronic game shown in FIG. 4.

[0017] FIG. 7 illustrates a second example screenshot and/or interface of the sequence that begins with FIG. 6.

[0018] FIG. 8 illustrates a third example screenshot and/or interface of the sequence that begins with FIG. 6.

[0019] FIG. 9 illustrates a fourth example screenshot and/or interface of the sequence that begins with FIG. 6.

[0020] FIG. 10 illustrates a fifth example screenshot and/or interface of the sequence that begins with FIG. 6.

[0021] FIG. 11 illustrates an exemplary method of electronic gaming with consolidating symbols, in accordance with the present disclosure.

DETAILED DESCRIPTION

[0022] Described herein are systems and methods for consolidating symbols in electronic gaming. For example, symbols may be consolidated in Class II or Class III games and/or primary/base games or bonus games. In the example embodiment, Wild symbols (e.g., symbols that may act as/be replaced by any other symbol in a game evaluation—for example, to achieve one or more high wins and/or allow winning combinations of symbols to be achieved more easily) are consolidated.

[0023] For example, Wild symbols may be displayed at any symbol position of a plurality of symbol positions. Winning paylines are paid based upon symbols displayed at the plurality of symbol positions. In the example embodiment, if two or more Wild symbols are displayed adjacent to each other, they consolidate into one Wild. Cascading symbols may fill in the symbol positions that the Wilds vacated, and any line wins are paid based upon the symbols displayed after the Wild symbols are consolidated. In some embodiments, any displayed Wild symbols may be consolidated (e.g., and do not need to be adjacent to each other to be consolidated). In some embodiments, a

number of displayed Wilds that are consolidated may be randomly determined.

[0024] Once a Wild is consolidated, the consolidated symbol may become one or more of a variety of different bonus options. For example, the new consolidated Wild may be displayed with a number associated therewith (e.g., a “Sticky Wild”). The number may represent how many spins the consolidated Wild will remain on the screen while other symbol positions are spun (e.g., resulting in a higher chance of winning during those spins because of the persistence of the Wild across a plurality of spins). As another example, the number associated with the Wild may indicate a multiplier to be applied to any win line pays (e.g., winning pay lines that include the Wild and/or that do not include the Wild). In some embodiments, the number may indicate a combination of the above examples (e.g., a number of spins and a multiplier) or any other bonus.

[0025] Certain technical benefits are realized based upon the present disclosure. For example, in Class II embodiments, scripts may be generated and stored in one or more lookup tables. RNG outcomes may determine which script to utilize to convey/communicate a game outcome (e.g., an outcome amount is determined and a random determination is made as to which symbol configurations/display sequences to display to communicate the outcome). For example, scripts may be stored according to output amount, and each script may include multiple game outcomes associated with an outcome amount (e.g., one or more of a preliminary outcome, an outcome as symbols are consolidating, and an outcome including a consolidated symbol). Further, because the scripts are associated with display sequences for different game outcomes, computational resources are saved and computer components operate more efficiently because, in some embodiments, once a script is determined, the script controls which display sequences to use to communicate a total game outcome (e.g., and no further determinations need to be made).

[0026] Further, in Class III embodiments additional/alternative technical benefits are realized. For example, a plurality of random determinations may be made (e.g., via a plurality of RNG outcomes and/or a plurality of lookup tables) in order to determine not only one or more game outcomes but also one or more display features. For example, which symbol position other symbols consolidate into may be randomly determined, a number of symbols to consolidate may be randomly determined, an order in which symbols consolidate into a consolidated symbol may be randomly determined, which bonus is associated with the consolidated symbol may be randomly determined, and so forth. The amount of random determinations possible in embodiments described herein (e.g., Class II, Class III, etc.) result in an increased variety of possible game outcomes to be provided and therefore an improved game as it is less likely any outcomes would be repeated (e.g., and/or displayed in the same manner).

[0027] Certain display benefits are also realized herein as technical benefits achieved by the present disclosure (e.g., because certain technical problems arise when trying to communicate/display a significant amount of information on a screen of limited size). For example, as explained herein, consolidating symbols are controlled to consolidate into a consolidated symbol (e.g., simultaneously, one-by-one, etc.). In some embodiments, animations may be displayed during the consolidation process to better communicate to a player that consolidation is occurring (e.g., a firework animation with each consolidation). In some embodiments, a number is incrementally increased as symbols consolidate (e.g., a “1” is displayed after a first consolidation, a “2” is displayed after a second consolidation, and so forth). The displayed number may be overlaid upon a consolidated symbol and/or associated with a bonus provided with a consolidated symbol (e.g., a number of free spins, a multiplier, etc.). Accordingly, the present disclosure provides a variety of improvements in communicating information to a player in a limited amount of display space/real estate—thereby providing an improved interface.

[0028] In some embodiments, consolidating symbols are consolidated in a determined order to maximize the number of consolidating symbols that are consolidated. For example, in some embodiments only adjacent consolidating symbols are consolidated. Accordingly, in those embodiments, consolidating symbols may be consolidated one-by-one (e.g., from left to right, right

to left, up to down, or down to up, as examples) such that consolidation of certain consolidating symbols does not leave vacant space adjacent another consolidating symbol, such that the consolidating symbol adjacent a vacant space may now not be consolidated because it is now adjacent a vacant space instead of a another consolidating symbol (e.g., see FIGS. 4 and 5 as an example).

[0029] To present further game outcomes once the consolidating symbols are consolidated, new symbols are added to now-vacant display positions (e.g., left vacant by the consolidating symbols). Accordingly, symbols may be displayed as “cascading” downward to fill in the vacant symbol positions left vacant from consolidating symbols, and also symbol positions left vacant by symbols that cascade downward to fill in previously-vacant symbol positions.

[0030] In the example embodiment, symbols are controlled as being consolidated and new symbols are added to vacant symbol positions all while controlling return to player (RTP). Certain technical problems are realized when generating new display improvements for electronic games because RTP must also be controlled. Accordingly, in example embodiments described herein, RTP is controlled by at least one lookup table (e.g., wherein at least one lookup in the at least one lookup table is controlled by a bingo outcome or an RNG call, as examples) indicating at least one of whether to consolidate consolidating symbols one by one or simultaneously, which bonus to apply to a consolidated symbol, which outcomes to provide while the consolidated symbol is present/displayed, and/or which symbols to add to the plurality of symbol positions to replace vacant symbol positions left behind by consolidating symbols and/or cascading symbols.

[0031] FIG. 1 illustrates several different models of EGMs which may be networked to various gaming related servers. Shown is a system **100** in a gaming environment including one or more server computers **102** (e.g., slot servers of a casino) that are in communication, via a communications network, with one or more gaming devices **104A-104X** (EGMs, slots, video poker, bingo machines, etc.) that can implement one or more aspects of the present disclosure. The gaming devices **104A-104X** may alternatively be portable and/or remote gaming devices such as, but not limited to, a smart phone, a tablet, a laptop, or a game console. Gaming devices **104A-104X** utilize specialized software and/or hardware to form non-generic, particular machines or apparatuses that comply with regulatory requirements regarding devices used for wagering or games of chance that provide monetary awards.

[0032] Communication between the gaming devices **104A-104X** and the server computers **102**, and among the gaming devices **104A-104X**, may be direct or indirect using one or more communication protocols. As an example, gaming devices **104A-104X** and the server computers **102** can communicate over one or more communication networks, such as over the Internet through a website maintained by a computer on a remote server or over an online data network including commercial online service providers, Internet service providers, private networks (e.g., local area networks and enterprise networks), and the like (e.g., wide area networks). The communication networks could allow gaming devices **104A-104X** to communicate with one another and/or the server computers **102** using a variety of communication-based technologies, such as radio frequency (RF) (e.g., wireless fidelity (WiFi®) and Bluetooth®), cable TV, satellite links and the like.

[0033] In some implementation, server computers **102** may not be necessary and/or preferred. For example, in one or more implementations, a stand-alone gaming device such as gaming device **104A**, gaming device **104B** or any of the other gaming devices **104C-104X** can implement one or more aspects of the present disclosure. However, it is typical to find multiple EGMs connected to networks implemented with one or more of the different server computers **102** described herein.

[0034] The server computers **102** may include a central determination gaming system server **106**, a ticket-in-ticket-out (TITO) system server **108**, a player tracking system server **110**, a progressive system server **112**, and/or a casino management system server **114**. Gaming devices **104A-104X** may include features to enable operation of any or all servers for use by the player and/or operator

(e.g., the casino, resort, gaming establishment, tavern, pub, etc.). For example, game outcomes may be generated on a central determination gaming system server **106** and then transmitted over the network to any of a group of remote terminals or remote gaming devices **104A-104X** that utilize the game outcomes and display the results to the players.

[0035] Gaming device **104A** is often of a cabinet construction which may be aligned in rows or banks of similar devices for placement and operation on a casino floor. The gaming device **104A** often includes a main door which provides access to the interior of the cabinet. Gaming device **104A** typically includes a button area or button deck **120** accessible by a player that is configured with input switches or buttons **122**, an access channel for a bill validator **124**, and/or an access channel for a ticket-out printer **126**.

[0036] In FIG. **1**, gaming device **104A** is shown as a Relm XL™ model gaming device manufactured by Aristocrat® Technologies, Inc. As shown, gaming device **104A** is a reel machine having a gaming display area **118** comprising a number (typically 3 or 5) of mechanical reels **130** with various symbols displayed on them. The mechanical reels **130** are independently spun and stopped to show a set of symbols within the gaming display area **118** which may be used to determine an outcome to the game.

[0037] In many configurations, the gaming device **104A** may have a main display **128** (e.g., video display monitor) mounted to, or above, the gaming display area **118**. The main display **128** can be a high-resolution liquid crystal display (LCD), plasma, light emitting diode (LED), or organic light emitting diode (OLED) panel which may be flat or curved as shown, a cathode ray tube, or other conventional electronically controlled video monitor.

[0038] In some implementations, the bill validator **124** may also function as a “ticket-in” reader that allows the player to use a casino issued credit ticket to load credits onto the gaming device **104A** (e.g., in a cashless ticket (“TITO”) system). In such cashless implementations, the gaming device **104A** may also include a “ticket-out” printer **126** for outputting a credit ticket when a “cash out” button is pressed. Cashless TITO systems are used to generate and track unique bar-codes or other indicators printed on tickets to allow players to avoid the use of bills and coins by loading credits using a ticket reader and cashing out credits using a ticket-out printer **126** on the gaming device **104A**. The gaming device **104A** can have hardware meters for purposes including ensuring regulatory compliance and monitoring the player credit balance. In addition, there can be additional meters that record the total amount of money wagered on the gaming device, total amount of money deposited, total amount of money withdrawn, total amount of winnings on gaming device **104A**.

[0039] In some implementations, a player tracking card reader **144**, a transceiver for wireless communication with a mobile device (e.g., a player's smartphone), a keypad **146**, and/or an illuminated display **148** for reading, receiving, entering, and/or displaying player tracking information is provided in gaming device **104A**. In such implementations, a game controller within the gaming device **104A** can communicate with the player tracking system server **110** to send and receive player tracking information.

[0040] Gaming device **104A** may also include a bonus topper wheel **134**. When bonus play is triggered (e.g., by a player achieving a particular outcome or set of outcomes in the primary game), bonus topper wheel **134** is operative to spin and stop with indicator arrow **136** indicating the outcome of the bonus game. Bonus topper wheel **134** is typically used to play a bonus game, but it could also be incorporated into play of the base or primary game.

[0041] A candle **138** may be mounted on the top of gaming device **104A** and may be activated by a player (e.g., using a switch or one of buttons **122**) to indicate to operations staff that gaming device **104A** has experienced a malfunction or the player requires service. The candle **138** is also often used to indicate a jackpot has been won and to alert staff that a hand payout of an award may be needed.

[0042] There may also be one or more information panels **152** which may be a back-lit,

silkscreened glass panel with lettering to indicate general game information including, for example, a game denomination (e.g., \$0.25 or \$1), pay lines, pay tables, and/or various game related graphics. In some implementations, the information panel(s) **152** may be implemented as an additional video display.

[0043] Gaming devices **104A** have traditionally also included a handle **132** typically mounted to the side of main cabinet **116** which may be used to initiate game play.

[0044] Many or all the above described components can be controlled by circuitry (e.g., a game controller) housed inside the main cabinet **116** of the gaming device **104A**, the details of which are shown in FIG. 2A.

[0045] An alternative example gaming device **104B** illustrated in FIG. 1 is the Arc™ model gaming device manufactured by Aristocrat® Technologies, Inc. Note that where possible, reference numerals identifying similar features of the gaming device **104A** implementation are also identified in the gaming device **104B** implementation using the same reference numbers. Gaming device **104B** does not include physical reels and instead shows game play functions on main display **128**. An optional topper screen **140** may be used as a secondary game display for bonus play, to show game features or attraction activities while a game is not in play, or any other information or media desired by the game designer or operator. In some implementations, the optional topper screen **140** may also or alternatively be used to display progressive jackpot prizes available to a player during play of gaming device **104B**.

[0046] Example gaming device **104B** includes a main cabinet **116** including a main door which opens to provide access to the interior of the gaming device **104B**. The main or service door is typically used by service personnel to refill the ticket-out printer **126** and collect bills and tickets inserted into the bill validator **124**. The main or service door may also be accessed to reset the machine, verify and/or upgrade the software, and for general maintenance operations.

[0047] Another example gaming device **104C** shown is the Helix™ model gaming device manufactured by Aristocrat® Technologies, Inc. Gaming device **104C** includes a main display **128A** that is in a landscape orientation. Although not illustrated by the front view provided, the main display **128A** may have a curvature radius from top to bottom, or alternatively from side to side. In some implementations, main display **128A** is a flat panel display. Main display **128A** is typically used for primary game play while secondary display **128B** is typically used for bonus game play, to show game features or attraction activities while the game is not in play or any other information or media desired by the game designer or operator. In some implementations, example gaming device **104C** may also include speakers **142** to output various audio such as game sound, background music, etc.

[0048] Many different types of games, including mechanical slot games, video slot games, video poker, video black jack, video pachinko, keno, bingo, and lottery, may be provided with or implemented within the depicted gaming devices **104A-104C** and other similar gaming devices. Each gaming device may also be operable to provide many different games. Games may be differentiated according to themes, sounds, graphics, type of game (e.g., slot game vs. card game vs. game with aspects of skill), denomination, number of paylines, maximum jackpot, progressive or non-progressive, bonus games, and may be deployed for operation in Class 2 or Class 3, etc.

[0049] FIG. 2A is a block diagram depicting exemplary internal electronic components of a gaming device **200** connected to various external systems. All or parts of the gaming device **200** shown could be used to implement any one of the example gaming devices **104A-X** depicted in FIG. 1. As shown in FIG. 2A, gaming device **200** includes a topper display **216** or another form of a top box (e.g., a topper wheel, a topper screen, etc.) that sits above cabinet **218**. Cabinet **218** or topper display **216** may also house a number of other components which may be used to add features to a game being played on gaming device **200**, including speakers **220**, a ticket printer **222** which prints bar-coded tickets or other media or mechanisms for storing or indicating a player's credit value, a ticket reader **224** which reads bar-coded tickets or other media or mechanisms for storing or

indicating a player's credit value, and a player tracking interface **232**. Player tracking interface **232** may include a keypad **226** for entering information, a player tracking display **228** for displaying information (e.g., an illuminated or video display), a card reader **230** for receiving data and/or communicating information to and from media or a device such as a smart phone enabling player tracking. FIG. 2 also depicts utilizing a ticket printer **222** to print tickets for a TITO system server **108**. Gaming device **200** may further include a bill validator **234**, player-input buttons **236** for player input, cabinet security sensors **238** to detect unauthorized opening of the cabinet **218**, a primary game display **240**, and a secondary game display **242**, each coupled to and operable under the control of game controller **202**.

[0050] The games available for play on the gaming device **200** are controlled by a game controller **202** that includes one or more processors **204**. Processor **204** represents a general-purpose processor, a specialized processor intended to perform certain functional tasks, or a combination thereof. As an example, processor **204** can be a central processing unit (CPU) that has one or more multi-core processing units and memory mediums (e.g., cache memory) that function as buffers and/or temporary storage for data. Alternatively, processor **204** can be a specialized processor, such as an application specific integrated circuit (ASIC), graphics processing unit (GPU), field-programmable gate array (FPGA), digital signal processor (DSP), or another type of hardware accelerator. In another example, processor **204** is a system on chip (SoC) that combines and integrates one or more general-purpose processors and/or one or more specialized processors. Although FIG. 2A illustrates that game controller **202** includes a single processor **204**, game controller **202** is not limited to this representation and instead can include multiple processors **204** (e.g., two or more processors).

[0051] FIG. 2A illustrates that processor **204** is operatively coupled to memory **208**. Memory **208** is defined herein as including volatile and nonvolatile memory and other types of non-transitory data storage components. Volatile memory is memory that do not retain data values upon loss of power. Nonvolatile memory is memory that do retain data values upon a loss of power. Examples of memory **208** include random access memory (RAM), read-only memory (ROM), hard disk drives, solid-state drives, universal serial bus (USB) flash drives, memory cards accessed via a memory card reader, floppy disks accessed via an associated floppy disk drive, optical discs accessed via an optical disc drive, magnetic tapes accessed via an appropriate tape drive, and/or other memory components, or a combination of any two or more of these memory components. In addition, examples of RAM include static random access memory (SRAM), dynamic random access memory (DRAM), magnetic random access memory (MRAM), and other such devices. Examples of ROM include a programmable read-only memory (PROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or other like memory device. Even though FIG. 2A illustrates that game controller **202** includes a single memory **208**, game controller **202** could include multiple memories **208** for storing program instructions and/or data.

[0052] Memory **208** can store one or more game programs **206** that provide program instructions and/or data for carrying out various implementations (e.g., game mechanics) described herein. Stated another way, game program **206** represents an executable program stored in any portion or component of memory **208**. In one or more implementations, game program **206** is embodied in the form of source code that includes human-readable statements written in a programming language or machine code that contains numerical instructions recognizable by a suitable execution system, such as a processor **204** in a game controller or other system. Examples of executable programs include: (1) a compiled program that can be translated into machine code in a format that can be loaded into a random access portion of memory **208** and run by processor **204**; (2) source code that may be expressed in proper format such as object code that is capable of being loaded into a random access portion of memory **208** and executed by processor **204**; and (3) source code that may be interpreted by another executable program to generate instructions in a random access

portion of memory **208** to be executed by processor **204**.

[0053] Alternatively, game programs **206** can be set up to generate one or more game instances based on instructions and/or data that gaming device **200** exchanges with one or more remote gaming devices, such as a central determination gaming system server **106** (not shown in FIG. 2A but shown in FIG. 1). For purpose of this disclosure, the term “game instance” refers to a play or a round of a game that gaming device **200** presents (e.g., via a user interface (UI)) to a player. The game instance is communicated to gaming device **200** via the network **214** and then displayed on gaming device **200**. For example, gaming device **200** may execute game program **206** as video streaming software that allows the game to be displayed on gaming device **200**. When a game is stored on gaming device **200**, it may be loaded from memory **208** (e.g., from a read only memory (ROM)) or from the central determination gaming system server **106** to memory **208**.

[0054] Gaming devices, such as gaming device **200**, are highly regulated to ensure fairness and, in many cases, gaming device **200** is operable to award monetary awards (e.g., typically dispensed in the form of a redeemable voucher). Therefore, to satisfy security and regulatory requirements in a gaming environment, hardware and software architectures are implemented in gaming devices **200** that differ significantly from those of general-purpose computers. Adapting general purpose computers to function as gaming devices **200** is not simple or straightforward because of: (1) the regulatory requirements for gaming devices **200**, (2) the harsh environment in which gaming devices **200** operate, (3) security requirements, (4) fault tolerance requirements, and (5) the requirement for additional special purpose componentry enabling functionality of an EGM. These differences require substantial engineering effort with respect to game design implementation, game mechanics, hardware components, and software.

[0055] One regulatory requirement for games running on gaming device **200** generally involves complying with a certain level of randomness. Typically, gaming jurisdictions mandate that gaming devices **200** satisfy a minimum level of randomness without specifying how a gaming device **200** should achieve this level of randomness. To comply, FIG. 2A illustrates that gaming device **200** could include an RNG **212** that utilizes hardware and/or software to generate RNG outcomes that lack any pattern. The RNG operations are often specialized and non-generic in order to comply with regulatory and gaming requirements. For example, in a slot game, game program **206** can initiate multiple RNG calls to RNG **212** to generate RNG outcomes, where each RNG call and RNG outcome corresponds to an outcome for a reel. In another example, gaming device **200** can be a Class II gaming device where RNG **212** generates RNG outcomes for creating Bingo cards. In one or more implementations, RNG **212** could be one of a set of RNGs operating on gaming device **200**. More generally, an output of the RNG **212** can be the basis on which game outcomes are determined by the game controller **202**. Game developers could vary the degree of true randomness for each RNG (e.g., pseudorandom) and utilize specific RNGs depending on game requirements. The output of the RNG **212** can include a random number or pseudorandom number (either is generally referred to as a “random number”).

[0056] In FIG. 2A, RNG **212** and hardware RNG **244** are shown in dashed lines to illustrate that RNG **212**, hardware RNG **244**, or both can be included in gaming device **200**. In one implementation, instead of including RNG **212**, gaming device **200** could include a hardware RNG **244** that generates RNG outcomes. Analogous to RNG **212**, hardware RNG **244** performs specialized and non-generic operations in order to comply with regulatory and gaming requirements. For example, because of regulation requirements, hardware RNG **244** could be a random number generator that securely produces random numbers for cryptography use. The gaming device **200** then uses the secure random numbers to generate game outcomes for one or more game features. In another implementation, the gaming device **200** could include both hardware RNG **244** and RNG **212**. RNG **212** may utilize the RNG outcomes from hardware RNG **244** as one of many sources of entropy for generating secure random numbers for the game features.

[0057] Another regulatory requirement for running games on gaming device **200** includes ensuring a certain level of RTP. Similar to the randomness requirement discussed above, numerous gaming jurisdictions also mandate that gaming device **200** provides a minimum level of RTP (e.g., RTP of at least 75%). A game can use one or more lookup tables (also called weighted tables) as part of a technical solution that satisfies regulatory requirements for randomness and RTP. In particular, a lookup table can integrate game features (e.g., trigger events for special modes or bonus games; newly introduced game elements such as extra reels, new symbols, or new cards; stop positions for dynamic game elements such as spinning reels, spinning wheels, or shifting reels; or card selections from a deck) with random numbers generated by one or more RNGs, so as to achieve a given level of volatility for a target level of RTP. (In general, volatility refers to the frequency or probability of an event such as a special mode, payout, etc. For example, for a target level of RTP, a higher-volatility game may have a lower payout most of the time with an occasional bonus having a very high payout, while a lower-volatility game has a steadier payout with more frequent bonuses of smaller amounts.) Configuring a lookup table can involve engineering decisions with respect to how RNG outcomes are mapped to game outcomes for a given game feature, while still satisfying regulatory requirements for RTP. Configuring a lookup table can also involve engineering decisions about whether different game features are combined in a given entry of the lookup table or split between different entries (for the respective game features), while still satisfying regulatory requirements for RTP and allowing for varying levels of game volatility.

[0058] FIG. 2A illustrates that gaming device **200** includes an RNG conversion engine **210** that translates the RNG outcome from RNG **212** to a game outcome presented to a player. To meet a designated RTP, a game developer can set up the RNG conversion engine **210** to utilize one or more lookup tables to translate the RNG outcome to a symbol element, stop position on a reel strip layout, and/or randomly chosen aspect of a game feature. As an example, the lookup tables can regulate a prize payout amount for each RNG outcome and how often the gaming device **200** pays out the prize payout amounts. The RNG conversion engine **210** could utilize one lookup table to map the RNG outcome to a game outcome displayed to a player and a second lookup table as a pay table for determining the prize payout amount for each game outcome. The mapping between the RNG outcome to the game outcome controls the frequency in hitting certain prize payout amounts.

[0059] FIG. 2A also depicts that gaming device **200** is connected over network **214** to player tracking system server **110**. Player tracking system server **110** may be, for example, an OASIS® system manufactured by Aristocrat® Technologies, Inc. Player tracking system server **110** is used to track play (e.g. amount wagered, games played, time of play and/or other quantitative or qualitative measures) for individual players so that an operator may reward players in a loyalty program. The player may use the player tracking interface **232** to access his/her account information, activate free play, and/or request various information. Player tracking or loyalty programs seek to reward players for their play and help build brand loyalty to the gaming establishment. The rewards typically correspond to the player's level of patronage (e.g., to the player's playing frequency and/or total amount of game plays at a given casino). Player tracking rewards may be complimentary and/or discounted meals, lodging, entertainment and/or additional play. Player tracking information may be combined with other information that is now readily obtainable by a casino management system.

[0060] When a player wishes to play the gaming device **200**, he/she can insert cash or a ticket voucher through a coin acceptor (not shown) or bill validator **234** to establish a credit balance on the gaming device. The credit balance is used by the player to place wagers on instances of the game and to receive credit awards based on the outcome of winning instances. The credit balance is decreased by the amount of each wager and increased upon a win. The player can add additional credits to the balance at any time. The player may also optionally insert a loyalty club card into the card reader **230**. During the game, the player views with one or more UIs, the game outcome on one or more of the primary game display **240** and secondary game display **242**. Other game and

prize information may also be displayed.

[0061] For each game instance, a player may make selections, which may affect play of the game. For example, the player may vary the total amount wagered by selecting the amount bet per line and the number of lines played. In many games, the player is asked to initiate or select options during course of game play (such as spinning a wheel to begin a bonus round or select various items during a feature game). The player may make these selections using the player-input buttons **236**, the primary game display **240** which may be a touch screen, or using some other device which enables a player to input information into the gaming device **200**.

[0062] During certain game events, the gaming device **200** may display visual and auditory effects that can be perceived by the player. These effects add to the excitement of a game, which makes a player more likely to enjoy the playing experience. Auditory effects include various sounds that are projected by the speakers **220**. Visual effects include flashing lights, strobing lights or other patterns displayed from lights on the gaming device **200** or from lights behind the information panel **152** (FIG. 1).

[0063] When the player is done, he/she cashes out the credit balance (typically by pressing a cash out button to receive a ticket from the ticket printer **222**). The ticket may be “cashed-in” for money or inserted into another machine to establish a credit balance for play.

[0064] Additionally, or alternatively, gaming devices **104A-104X** and **200** can include or be coupled to one or more wireless transmitters, receivers, and/or transceivers (not shown in FIGS. 1 and 2A) that communicate (e.g., Bluetooth® or other near-field communication technology) with one or more mobile devices to perform a variety of wireless operations in a casino environment. Examples of wireless operations in a casino environment include detecting the presence of mobile devices, performing credit, points, comps, or other marketing or hard currency transfers, establishing wagering sessions, and/or providing a personalized casino-based experience using a mobile application. In one implementation, to perform these wireless operations, a wireless transmitter or transceiver initiates a secure wireless connection between a gaming device **104A-104X** and **200** and a mobile device. After establishing a secure wireless connection between the gaming device **104A-104X** and **200** and the mobile device, the wireless transmitter or transceiver does not send and/or receive application data to and/or from the mobile device. Rather, the mobile device communicates with gaming devices **104A-104X** and **200** using another wireless connection (e.g., WiFi® or cellular network). In another implementation, a wireless transceiver establishes a secure connection to directly communicate with the mobile device. The mobile device and gaming device **104A-104X** and **200** sends and receives data utilizing the wireless transceiver instead of utilizing an external network. For example, the mobile device would perform digital wallet transactions by directly communicating with the wireless transceiver. In one or more implementations, a wireless transmitter could broadcast data received by one or more mobile devices without establishing a pairing connection with the mobile devices.

[0065] Although FIGS. 1 and 2A illustrate specific implementations of a gaming device (e.g., gaming devices **104A-104X** and **200**), the disclosure is not limited to those implementations shown in FIGS. 1 and 2. For example, not all gaming devices suitable for implementing implementations of the present disclosure necessarily include top wheels, top boxes, information panels, cashless ticket systems, and/or player tracking systems. Further, some suitable gaming devices have only a single game display that includes only a mechanical set of reels and/or a video display, while others are designed for bar counters or tabletops and have displays that face upwards. Gaming devices **104A-104X** and **200** may also include other processors that are not separately shown. Using FIG. 2A as an example, gaming device **200** could include display controllers (not shown in FIG. 2A) configured to receive video input signals or instructions to display images on game displays **240** and **242**. Alternatively, such display controllers may be integrated into the game controller **202**. The use and discussion of FIGS. 1 and 2 are examples to facilitate ease of description and explanation.

[0066] FIG. 2B depicts a casino gaming environment according to one example. In this example,

the casino **251** includes banks **252** of EGMs **104**. In this example, each bank **252** of EGMs **104** includes a corresponding gaming signage system **254** (also shown in FIG. 2A). According to this implementation, the casino **251** also includes mobile gaming devices **256**, which are also configured to present wagering games in this example. The mobile gaming devices **256** may, for example, include tablet devices, cellular phones, smart phones and/or other handheld devices. In this example, the mobile gaming devices **256** are configured for communication with one or more other devices in the casino **251**, including but not limited to one or more of the server computers **102**, via wireless access points **258**.

[0067] According to some examples, the mobile gaming devices **256** may be configured for stand-alone determination of game outcomes. However, in some alternative implementations the mobile gaming devices **256** may be configured to receive game outcomes from another device, such as the central determination gaming system server **106**, one of the EGMs **104**, etc.

[0068] Some mobile gaming devices **256** may be configured to accept monetary credits from a credit or debit card, via a wireless interface (e.g., via a wireless payment app), via tickets, via a patron casino account, etc. However, some mobile gaming devices **256** may not be configured to accept monetary credits via a credit or debit card. Some mobile gaming devices **256** may include a ticket reader and/or a ticket printer whereas some mobile gaming devices **256** may not, depending on the particular implementation.

[0069] In some implementations, the casino **251** may include one or more kiosks **260** that are configured to facilitate monetary transactions involving the mobile gaming devices **256**, which may include cash out and/or cash in transactions. The kiosks **260** may be configured for wired and/or wireless communication with the mobile gaming devices **256**. The kiosks **260** may be configured to accept monetary credits from casino patrons **262** and/or to dispense monetary credits to casino patrons **262** via cash, a credit or debit card, via a wireless interface (e.g., via a wireless payment app), via tickets, etc. According to some examples, the kiosks **260** may be configured to accept monetary credits from a casino patron and to provide a corresponding amount of monetary credits to a mobile gaming device **256** for wagering purposes, e.g., via a wireless link such as a near-field communications link. In some such examples, when a casino patron **262** is ready to cash out, the casino patron **262** may select a cash out option provided by a mobile gaming device **256**, which may include a real button or a virtual button (e.g., a button provided via a graphical user interface) in some instances. In some such examples, the mobile gaming device **256** may send a “cash out” signal to a kiosk **260** via a wireless link in response to receiving a “cash out” indication from a casino patron. The kiosk **260** may provide monetary credits to the casino patron **262** corresponding to the “cash out” signal, which may be in the form of cash, a credit ticket, a credit transmitted to a financial account corresponding to the casino patron, etc.

[0070] In some implementations, a cash-in process and/or a cash-out process may be facilitated by the TITO system server **108**. For example, the TITO system server **108** may control, or at least authorize, ticket-in and ticket-out transactions that involve a mobile gaming device **256** and/or a kiosk **260**.

[0071] Some mobile gaming devices **256** may be configured for receiving and/or transmitting player loyalty information. For example, some mobile gaming devices **256** may be configured for wireless communication with the player tracking system server **110**. Some mobile gaming devices **256** may be configured for receiving and/or transmitting player loyalty information via wireless communication with a patron's player loyalty card, a patron's smartphone, etc.

[0072] According to some implementations, a mobile gaming device **256** may be configured to provide safeguards that prevent the mobile gaming device **256** from being used by an unauthorized person. For example, some mobile gaming devices **256** may include one or more biometric sensors and may be configured to receive input via the biometric sensor(s) to verify the identity of an authorized patron. Some mobile gaming devices **256** may be configured to function only within a predetermined or configurable area, such as a casino gaming area.

[0073] FIG. 2C is a diagram that shows examples of components of a system for providing online gaming according to some aspects of the present disclosure. As with other figures presented in this disclosure, the numbers, types and arrangements of gaming devices shown in FIG. 2C are merely shown by way of example. In this example, various gaming devices, including but not limited to end user devices (EUDs) **264a**, **264b** and **264c** are capable of communication via one or more networks **417**. The networks **417** may, for example, include one or more cellular telephone networks, the Internet, etc. In this example, the EUDs **264a** and **264b** are mobile devices: according to this example the EUD **264a** is a tablet device and the EUD **264b** is a smart phone. In this implementation, the EUD **264c** is a laptop computer that is located within a residence **266** at the time depicted in FIG. 2C. Accordingly, in this example the hardware of EUDs is not specifically configured for online gaming, although each EUD is configured with software for online gaming. For example, each EUD may be configured with a web browser. Other implementations may include other types of EUD, some of which may be specifically configured for online gaming.

[0074] In this example, a gaming data center **276** includes various devices that are configured to provide online wagering games via the networks **417**. The gaming data center **276** is capable of communication with the networks **417** via the gateway **272**. In this example, switches **278** and routers **280** are configured to provide network connectivity for devices of the gaming data center **276**, including storage devices **282a**, servers **284a** and one or more workstations **286a**. The servers **284a** may, for example, be configured to provide access to a library of games for online game play. In some examples, code for executing at least some of the games may initially be stored on one or more of the storage devices **282a**. The code may be subsequently loaded onto a server **284a** after selection by a player via an EUD and communication of that selection from the EUD via the networks **417**. The server **284a** onto which code for the selected game has been loaded may provide the game according to selections made by a player and indicated via the player's EUD. In other examples, code for executing at least some of the games may initially be stored on one or more of the servers **284a**. Although only one gaming data center **276** is shown in FIG. 2C, some implementations may include multiple gaming data centers **276**.

[0075] In this example, a financial institution data center **270** is also configured for communication via the networks **417**. Here, the financial institution data center **270** includes servers **284b**, storage devices **282b**, and one or more workstations **286b**. According to this example, the financial institution data center **270** is configured to maintain financial accounts, such as checking accounts, savings accounts, loan accounts, etc. In some implementations one or more of the authorized users **274a-274c** may maintain at least one financial account with the financial institution that is serviced via the financial institution data center **270**.

[0076] According to some implementations, the gaming data center **276** may be configured to provide online wagering games in which money may be won or lost. According to some such implementations, one or more of the servers **284a** may be configured to monitor player credit balances, which may be expressed in game credits, in currency units, or in any other appropriate manner. In some implementations, the server(s) **284a** may be configured to obtain financial credits from and/or provide financial credits to one or more financial institutions, according to a player's "cash in" selections, wagering game results and a player's "cash out" instructions. According to some such implementations, the server(s) **284a** may be configured to electronically credit or debit the account of a player that is maintained by a financial institution, e.g., an account that is maintained via the financial institution data center **270**. The server(s) **284a** may, in some examples, be configured to maintain an audit record of such transactions.

[0077] In some alternative implementations, the gaming data center **276** may be configured to provide online wagering games for which credits may not be exchanged for cash or the equivalent. In some such examples, players may purchase game credits for online game play, but may not "cash out" for monetary credit after a gaming session. Moreover, although the financial institution data center **270** and the gaming data center **276** include their own servers and storage devices in

this example, in some examples the financial institution data center **270** and/or the gaming data center **276** may use offsite “cloud-based” servers and/or storage devices. In some alternative examples, the financial institution data center **270** and/or the gaming data center **276** may rely entirely on cloud-based servers.

[0078] One or more types of devices in the gaming data center **276** (or elsewhere) may be capable of executing middleware, e.g., for data management and/or device communication. Authentication information, player tracking information, etc., including but not limited to information obtained by EUDs **264** and/or other information regarding authorized users of EUDs **264** (including but not limited to the authorized users **274a-274c**), may be stored on storage devices **282** and/or servers **284**. Other game-related information and/or software, such as information and/or software relating to leaderboards, players currently playing a game, game themes, game-related promotions, game competitions, etc., also may be stored on storage devices **282** and/or servers **284**. In some implementations, some such game-related software may be available as “apps” and may be downloadable (e.g., from the gaming data center **276**) by authorized users.

[0079] In some examples, authorized users and/or entities (such as representatives of gaming regulatory authorities) may obtain gaming-related information via the gaming data center **276**. One or more other devices (such EUDs **264** or devices of the gaming data center **276**) may act as intermediaries for such data feeds. Such devices may, for example, be capable of applying data filtering algorithms, executing data summary and/or analysis software, etc. In some implementations, data filtering, summary and/or analysis software may be available as “apps” and downloadable by authorized users.

[0080] FIG. **3** illustrates, in block diagram form, an implementation of a game processing architecture **300** that implements a game processing pipeline for the play of a game in accordance with various implementations described herein. As shown in FIG. **3**, the gaming processing pipeline starts with having a UI system **302** receive one or more player inputs for the game instance. Based on the player input(s), the UI system **302** generates and sends one or more RNG calls to a game processing backend system **314**. Game processing backend system **314** then processes the RNG calls with RNG engine **316** to generate one or more RNG outcomes. The RNG outcomes are then sent to the RNG conversion engine **320** to generate one or more game outcomes for the UI system **302** to display to a player. The game processing architecture **300** can implement the game processing pipeline using a gaming device, such as gaming devices **104A-104X** and **200** shown in FIGS. **1** and **2**, respectively. Alternatively, portions of the gaming processing architecture **300** can implement the game processing pipeline using a gaming device and one or more remote gaming devices, such as central determination gaming system server **106** shown in FIG. **1**.

[0081] The UI system **302** includes one or more UIs that a player can interact with. The UI system **302** could include one or more game play UIs **304**, one or more bonus game play UIs **308**, and one or more multiplayer UIs **312**, where each UI type includes one or more mechanical UIs and/or graphical UIs (GUIs). In other words, game play UI **304**, bonus game play UI **308**, and the multiplayer UI **312** may utilize a variety of UI elements, such as mechanical UI elements (e.g., physical “spin” button or mechanical reels) and/or GUI elements (e.g., virtual reels shown on a video display or a virtual button deck) to receive player inputs and/or present game play to a player. Using FIG. **3** as an example, the different UI elements are shown as game play UI elements **306A-306N** and bonus game play UI elements **310A-310N**.

[0082] The game play UI **304** represents a UI that a player typically interfaces with for a base game. During a game instance of a base game, the game play UI elements **306A-306N** (e.g., GUI elements depicting one or more virtual reels) are shown and/or made available to a user. In a subsequent game instance, the UI system **302** could transition out of the base game to one or more bonus games. The bonus game play UI **308** represents a UI that utilizes bonus game play UI elements **310A-310N** for a player to interact with and/or view during a bonus game. In one or more implementations, at least some of the game play UI element **306A-306N** are similar to the bonus

game play UI elements **310A-310N**. In other implementations, the game play UI element **306A-306N** can differ from the bonus game play UI elements **310A-310N**.

[0083] FIG. 3 also illustrates that UI system **302** could include a multiplayer UI **312** purposed for game play that differs or is separate from the typical base game. For example, multiplayer UI **312** could be set up to receive player inputs and/or presents game play information relating to a tournament mode. When a gaming device transitions from a primary game mode that presents the base game to a tournament mode, a single gaming device is linked and synchronized to other gaming devices to generate a tournament outcome. For example, multiple RNG engines **316** corresponding to each gaming device could be collectively linked to determine a tournament outcome. To enhance a player's gaming experience, tournament mode can modify and synchronize sound, music, reel spin speed, and/or other operations of the gaming devices according to the tournament game play. After tournament game play ends, operators can switch back the gaming device from tournament mode to a primary game mode to present the base game. Although FIG. 3 does not explicitly depict that multiplayer UI **312** includes UI elements, multiplayer UI **312** could also include one or more multiplayer UI elements.

[0084] Based on the player inputs, the UI system **302** could generate RNG calls to a game processing backend system **314**. As an example, the UI system **302** could use one or more application programming interfaces (APIs) to generate the RNG calls. To process the RNG calls, the RNG engine **316** could utilize gaming RNG **318** and/or non-gaming RNGs **319A-319N**. Gaming RNG **318** could correspond to RNG **212** or hardware RNG **244** shown in FIG. 2A. As previously discussed with reference to FIG. 2A, gaming RNG **318** often performs specialized and non-generic operations that comply with regulatory and/or game requirements. For example, because of regulation requirements, gaming RNG **318** could correspond to RNG **212** by being a cryptographic RNG or pseudorandom number generator (PRNG) (e.g., Fortuna PRNG) that securely produces random numbers for one or more game features. To securely generate random numbers, gaming RNG **318** could collect random data from various sources of entropy, such as from an operating system (OS) and/or a hardware RNG (e.g., hardware RNG **244** shown in FIG. 2A). Alternatively, non-gaming RNGs **319A-319N** may not be cryptographically secure and/or be computationally less expensive. Non-gaming RNGs **319A-319N** can, thus, be used to generate outcomes for non-gaming purposes. As an example, non-gaming RNGs **319A-319N** can generate random numbers for generating random messages that appear on the gaming device.

[0085] The RNG conversion engine **320** processes each RNG outcome from RNG engine **316** and converts the RNG outcome to a UI outcome that is feedback to the UI system **302**. With reference to FIG. 2A, RNG conversion engine **320** corresponds to RNG conversion engine **210** used for game play. As previously described, RNG conversion engine **320** translates the RNG outcome from the RNG **212** to a game outcome presented to a player. RNG conversion engine **320** utilizes one or more lookup tables **322A-322N** to regulate a prize payout amount for each RNG outcome and how often the gaming device pays out the derived prize payout amounts. In one example, the RNG conversion engine **320** could utilize one lookup table to map the RNG outcome to a game outcome displayed to a player and a second lookup table as a pay table for determining the prize payout amount for each game outcome. In this example, the mapping between the RNG outcome and the game outcome controls the frequency in hitting certain prize payout amounts. Different lookup tables could be utilized depending on the different game modes, for example, a base game versus a bonus game.

[0086] After generating the UI outcome, the game processing backend system **314** sends the UI outcome to the UI system **302**. Examples of UI outcomes are symbols to display on a video reel or reel stops for a mechanical reel. In one example, if the UI outcome is for a base game, the UI system **302** updates one or more game play UI elements **306A-306N**, such as symbols, for the game play UI **304**. In another example, if the UI outcome is for a bonus game, the UI system could update one or more bonus game play UI elements **310A-310N** (e.g., symbols) for the bonus game

play UI **308**. In response to updating the appropriate UI, the player may subsequently provide additional player inputs to initiate a subsequent game instance that progresses through the game processing pipeline.

[0087] FIG. **4** illustrates an example screenshot **400** and/or interface of an electronic game before symbols are consolidated, in accordance with the present disclosure. As shown in FIG. **4**, a plurality of symbols **404** are displayed in a display area **402** of limited size (e.g., of a display device) at a plurality of symbol positions **406**. Plurality of symbols **404** includes four Wild symbols **408A-D**. In some embodiments, plurality of symbols **404** may be determined by a bingo outcome (e.g., Class II embodiments). In some embodiments, plurality of symbols **404** may be determined by an RNG call (e.g., to RNG **318**) and one or more lookup tables (e.g., **322A-N**).

[0088] As shown by arrows **410** in FIG. **4**, Wild symbols **408A-D** will be consolidated into a consolidated Wild symbol (shown in FIG. **5**). In the example embodiment, before Wild symbols **408A-D** are consolidated, any winning paylines associated with plurality of symbols **404** are paid/outputted.

[0089] It should be appreciated that while Wild symbols **408A-D** are described with respect to an example embodiment herein, it is envisioned that any type of symbol(s) may be consolidated. Further, symbols may be consolidated in a variety of ways in accordance with the present disclosure. For example, in some embodiments Wilds may be consolidated simultaneously. In some embodiments, Wilds may be consolidated one-by-one (e.g., or by twos, threes, etc.). In embodiments where Wilds are consolidated one at a time, payouts may be made each time a Wild is consolidated. In other words, a first payout would be made based upon an initial plurality of symbols shown, a second payout would be made after a first Wild was consolidated, a third payout would be made after a second Wild was consolidated, and so forth (e.g., with symbols replacing vacant spaces before each payout). In some embodiments, the order in which Wilds are consolidated may be determined/configured such that the payouts after each respective consolidation increase with respect to previous payouts (e.g., to provide a better gameplay experience).

[0090] Further, symbol positions vacated by symbols that are consolidated may be filled with new symbols in a variety of ways. For example, vacated symbol positions may each be their own reel that is spun again after a symbol previously displayed therein is consolidated (e.g., such that a new symbol is displayed therein once the reel is stopped). In some embodiments, symbols may “cascade” downward to replace vacated symbol positions (e.g., see FIG. **5** as an example). In some embodiments, vacant spaces may all be filled by the same symbol (e.g., a scatter symbol) and/or different symbols (e.g., randomly determined).

[0091] In some embodiments, the symbol to which other symbols are consolidated may be randomly determined. In some embodiments, the symbol to which other symbols are consolidated may be predetermined (e.g., according to a lookup table).

[0092] Symbols may be consolidated in Class II or Class III games and/or primary/base games or bonus games. In the example embodiment, Wild symbols (e.g., symbols that may act as/be replaced by any other symbol in a game evaluation—for example, to achieve one or more high wins and/or allow winning combinations of symbols to be achieved more easily) are consolidated.

[0093] For example, winning paylines are paid based upon symbols displayed at plurality of symbol positions **406**. In the example embodiment, symbols **408A-D** consolidate into one Wild. Cascading symbols may fill in the symbol positions that Wilds **408A-D** vacated, and any line wins are paid based upon the symbols displayed after the Wild symbols are consolidated.

[0094] Once a Wild is consolidated, it may become one or more of a variety of different options (e.g., see FIG. **5**). For example, the new consolidated Wild may be displayed with a number associated therewith (e.g., a “Sticky Wild”). The number may represent how many spins the consolidated Wild will remain on the screen (e.g., resulting in a higher chance of winning during those spins because of the persistence of the Wild across a plurality of spins). As another example,

the number associated with the Wild may indicate a multiplier to be applied to any win line pays (e.g., win lines that include the Wild and/or that do not include the Wild). In some embodiments, the number may indicate a combination of the above examples (e.g., a number of spins and a multiplier) or any other bonus.

[0095] FIG. 5 illustrates an example screenshot **500** and/or interface of the electronic game shown in FIG. 4 after symbols are consolidated. As shown in FIG. 5, symbols **408A-D** in display area **402** have been consolidated into a consolidated symbol **502**. Further, vacant symbol positions left by symbols **408A-D** have been filled by “cascading” (e.g., slid downward) symbols from above. For example, Mt. Olympus symbol **504** has slid downward to replace symbol **408D** and 10 symbol **506** has slid down to replace symbol **408B**. Symbol **408C** was positioned in an upper row of symbol positions **406** and thus no symbol was available to cascade downward. Accordingly, a new Scatter symbol **508** has replaced symbol **408C**. As symbols **504** and **506** cascaded downward, new vacant symbol positions were created. Accordingly, 9 symbol **510** slid down one position and a new 10 symbol **512** replaced the position previously occupied by 9 symbol **510**. A new ship symbol **514** has replaced symbol **408B**.

[0096] Further, consolidated symbol **502** includes an indicator **516** currently displaying the number 3. In the example shown in FIG. 5, indicator **516** indicates a number of spins (e.g., free spins) for which consolidated symbol **502** will remain in display area **402**. Accordingly, 3 spins including consolidated symbol **502** will occur (e.g., all other symbol positions/reels will be spun other than the symbol position including symbol **502** for 3 spins). In the example embodiment, indicator **516** is based upon the number of symbols consolidated into consolidated symbol **502**. For example, as shown in FIG. 5, 3 symbols (**408B-D**) were consolidated (e.g., into symbol **408A**), and thus 3 spins are presented. With each spin of the electronic game including indicator **516**, indicator may be decremented until 0 spins remain and symbol **502** is no longer present.

[0097] In some embodiments, indicator **516** may indicate a bonus other than a number of spins and may be based on factors other than an amount of symbols consolidated into consolidated symbol **502**. For example, indicator **516** may indicate an award multiplier to be applied to a number of paylines (e.g., paylines including and/or excluding consolidated symbol **502**).

[0098] In some embodiments, only consolidating symbols **408A-D** that are adjacent another consolidating symbol **408A-D** in the plurality of symbols are consolidated. In some embodiments, any displayed consolidating symbols **408A-D** are consolidated.

[0099] In embodiments where adjacent consolidating symbols are consolidated, consolidating symbols may be consolidated one-by-one (e.g., from left to right, right to left, up to down, or down to up, as examples) such that consolidation of certain consolidating symbols does not leave vacant space adjacent another consolidating symbol, such that the consolidating symbol adjacent a vacant space may now not be consolidated because it is now adjacent a vacant space instead of a another consolidating symbol. In the example of FIGS. 4 and 5, for example, symbol **408B** would not be first controlled as consolidating to the symbol position of symbol **408A** because that would leave the symbol position previously occupied by symbol **408B** vacant (e.g., not including a consolidating symbol). Thus, symbol **408C** would not be adjacent a consolidating symbol and would not be consolidated. Accordingly, to maximize the number of adjacent symbols consolidated, symbol **408C** would first be controlled as consolidating to the position of symbol **408B**, and then a consolidation would occur to the position of symbol **408A**, resulting in display of consolidated symbol **502** (e.g., in other embodiments symbols **408A-C** may be consolidated left to right, resulting in consolidated symbol **502** being displayed at the symbol position previously occupied by symbol **408C**).

[0100] As explained herein, game outcomes may be provided at various moments before, during, and after consolidation of consolidating symbols. For example, a first outcome may be provided when a first plurality of symbols are displayed (e.g., as shown in FIG. 4). Other outcomes may be provided as consolidating symbols consolidate. For example, outcomes may be provided as

symbols consolidated one-by-one (e.g., in the example shown in FIGS. 4 and 5, three outcomes may be provided during consolidation, one after each of symbols **408B-D** consolidate to the position of **408A** and vacant positions left by symbols **408B-D** are replaced) and/or symbols may consolidate all at once (e.g., in the example shown in FIGS. 4 and 5, one outcome may be provided during consolidation after symbols **408B-D** consolidate to the position of **408A** simultaneously and vacant positions left by symbols **408B-D** are replaced). Further outcomes may be provided while a consolidated symbol is present. For example, a consolidated symbol may be present for three spins, as shown in FIG. 5.

[0101] Game outcomes as described herein may be selected and/or determined to control RTP all while providing the various other benefits (e.g., display benefits) described herein. For example, in some embodiments, a total game outcome may be determined for the electronic game. The total game outcome may be utilized to determine any outcomes provided based upon the initial plurality of symbols (e.g., shown in FIG. 4), outcomes provided during consolidation, and/or outcomes provided while the consolidated symbol is present. For example, a total outcome of 1000 credits may initially be determined. Accordingly, the total outcome of any outcomes provided based upon the initial plurality of symbols, outcomes provided during consolidation, and/or outcomes provided while the consolidated symbol is present would need to total 1000 credits (e.g., and the systems and methods described herein control at least the outcomes described above such that the total of the outcomes would equal 1000 credits).

[0102] Splitting of a total game outcome (e.g., based upon a bingo outcome, lookup table, and or RNG call) amongst different outcomes is important for controlling RTP while managing computer resources. For example, generating a total outcome before generating other outcomes (e.g., to total the total outcome) limits the number of outcomes that could be presented based upon the initial plurality of symbols, during consolidation, and/or while the consolidated symbol is present. In some embodiments, the outcomes whose summation equals the total outcome may be presented in order of equal and/or increasing output amount. In other words, the initial plurality of symbols may be associated with a lower output amount, outcomes presented during consolidation may be presented in order of equal and/or increasing output, and outcomes presented while the consolidated symbol is present may also be presented in order of equal and/or increasing output (e.g., wherein the initial output is a lowest output, the outputs during consolidation are equal to or greater than the initial output, and the outputs while the consolidated symbol is present are equal to or greater than the outputs during consolidation).

[0103] FIG. 6 illustrates a first example screenshot **600** and/or interface of a sequence of the electronic game shown in FIG. 4. As shown in FIG. 6, a plurality of symbols **604** are displayed in a display area **602** at a plurality of symbols positions **606**. Plurality of symbols **604** includes three Wild symbols **608A-C**.

[0104] As explained herein, symbols **604** communicate a game outcome. Accordingly, a win associated with symbols **604** is displayed at win display area **610**. A bet amount is shown at bet amount display area **612** and a total credit amount (e.g., for the gaming session and associated with a player account) is displayed at credit display area **614**. In the example shown in FIG. 6, symbols **604** are associated with a two hundred credit win. Thus, two hundred credits are displayed at win display area **610**. For example, the two hundred credit win may be associated with a subset **616** of symbols **604** across a payline **618**. To illustrate the subset **616**, each symbol position of the subset may include a highlighted border **620** (e.g., or other animation).

[0105] FIG. 7 illustrates a second example screenshot **700** and/or interface of the sequence that begins with FIG. 6. Wild symbols **608A-C** are determined to be consolidated, as described herein (e.g., and as illustrated by arrows **702, 704**). Notably, symbols **608A-C** are consolidated after the game outcome (e.g., two hundred credits) associated with evaluation of symbols **604** is presented.

[0106] FIG. 8 illustrates a third example screenshot **800** and/or interface of the sequence that begins with FIG. 6. As shown in FIG. 8, Wild symbols **608A-C** have been consolidated into

consolidated Wild symbol **802**. Notably, win display area **610**, bet amount display area **612**, and credit display area **614** display the same amounts as shown in FIG. **6** because no additional spins have occurred.

[0107] For example, three Wild symbols **608A-C** were consolidated. Accordingly, an x3 icon **804** is displayed on (e.g., overlaid on) symbol **802**. In some embodiments, x3 icon **804** may correspond to an award multiplier. For example, a win amount involving consolidated Wild symbol **802** with x3 icon **804** may be multiplied by three (or any other value, as indicated by the numeral following, or in some instances preceding, the “x” of icon **804**, corresponding to the number of Wilds consolidated).

[0108] In some examples, consolidated Wild symbol **802** may be a persistent (e.g., “sticky”) Wild symbol with x3 icon **804** corresponding to a number of subsequent reel spins in which the sticky Wild symbol will remain in the current symbol position (e.g., row 2, column 2). For example, as shown in FIG. **8**, x3 icon **804** corresponds to three consolidated Wilds (e.g., **608A-C**). In other examples, x3 icon **804** may be a value other than three, as indicated by the numeral following, or in some instances preceding, the “x” of icon **804** being different from three and corresponding to the number of Wilds consolidated.

[0109] Further, in some instances, icon **804** may decrement with each successive reel spin, finally ‘spinning away’ when the numeral of icon **804** decrements to 0. In this example, the numeral following (or preceding) the “x” of icon **804** may correspond to both an award multiplier and a number of persistent spins indicator, with the value decrementing with each successive spin.

[0110] In FIG. **8**, Wild symbols **608B**, **608C** have vacated their respective symbol positions, and cascading symbols have replaced Wild symbols **608B**, **608C**. For instance, the “10” symbol has “cascaded” down one symbol position to fill the symbol position left vacant by Wild symbol **608B** consolidating, and a new symbol (e.g., a unicorn as shown) has replaced the symbol position that the “10” symbol vacated. Further, a new symbol (e.g., a “K” or King symbol as shown) has replaced the symbol position left vacant by Wild symbol **608C** consolidating. Down arrows **806**, **808** illustrate the cascading described above.

[0111] FIG. **9** illustrates a fourth example screenshot **900** and/or interface of the sequence that begins with FIG. **6**. As shown in FIG. **9**, a next game outcome is presented after the consolidation of Wild symbols **608A-C** into consolidated symbol **802**. For example, plurality of symbols **902** are associated with a win of three hundred credits. FIG. **9** shows that the three hundred credit win, shown in win display area **610**, is associated with a subset **904** of the plurality of symbols **902**, including consolidated symbol **802**. In this example, the three hundred credit win corresponds to a three-symbol combination (two Aces and a Wild) corresponding to a one hundred credit win multiplied by a times three multiplier (indicated by the “3” of x3 icon **804**). To illustrate the subset **616**, each symbol position of the subset may include a highlighted border **906** (e.g., or other animation). Credit display area **614** is also increased by three hundred credits (e.g., according to the next game outcome).

[0112] FIG. **10** illustrates a fifth example screenshot **1000** and/or interface of the sequence that begins with FIG. **6**. As shown in FIG. **10**, a next game outcome is presented while consolidated symbol **802** is held in place (e.g., for a first of three spins where consolidated symbol **802** will be held, based on x3 icon **804**). As a next game outcome has been presented while consolidated symbol **802** is held, x3 icon **804** is decremented by one to an x2 icon **1002** on consolidated symbol **802** to communicate that consolidated symbol **802** will be held for two additional spins.

[0113] A new plurality of symbols **1004** illustrated in FIG. **10**, and subset thereof, are associated with a fourteen hundred credit win, as shown in win display area **610** (e.g., and as incremented in credit display area **614**). For example, fourteen hundred credits are associated with two payline wins **1006** (corresponding to a three-symbol combination, one hundred credit win), and **1008** (corresponding to a four-symbol combination, six hundred credit win) multiplied by a times two multiplier (indicated by the “2” of x2 icon **1002**) as shown. To illustrate the subset, each symbol

position of the subset may include a highlighted border **1010** (e.g., or other animation).

[0114] FIGS. **6-10** illustrate, in some embodiments, the splitting of a total game outcome (e.g., based upon a bingo outcome, lookup table, and or RNG call) amongst different outcomes (e.g., in a Class II embodiment). For example, a total outcome may be generated (e.g., including the two hundred credit, three hundred credit, and fourteen hundred credit wins) before generating other outcomes (e.g., to total the total outcome). As shown in FIGS. **6-10**, the outcomes whose summation equals the total outcome are presented in order of increasing output amount. Accordingly, the two hundred credit outcome is presented first, followed by the three hundred credit outcome, and then the fourteen hundred credit outcome.

[0115] In other words, in Class II embodiments, scripts may be generated and stored in one or more lookup tables. RNG outcomes may determine which script to utilize to convey/communicate a total game outcome (e.g., an outcome amount is determined and a random determination is made as to which symbol configurations/display sequences to display to communicate the outcome). FIGS. **6-10** illustrate a portion of one such sequence (e.g., that may include sorting (e.g., increasing and/or decreasing) and/or filtering (e.g., removing losing game outcomes) game outcomes based on output amount). Because the scripts are associated with display sequences for different game outcomes, computational resources are saved and computer components operate more efficiently because, in some embodiments, once a script is determined, the script controls which display sequences to use to communicate a total game outcome (e.g., and no further determinations need to be made). For instance, in some embodiments of FIGS. **6-10**, the outcomes presented in FIGS. **7-10** are determined when the outcome shown in FIG. **6** is determined (e.g., to allow for determining which outcomes to utilize to present a total game outcome, and/or sorting and/or filtering those outcomes).

[0116] FIG. **11** illustrates an exemplary method **1100** of electronic gaming with consolidating symbols, in accordance with the present disclosure. In the example embodiment, method **1100** includes causing display **1102** of a plurality of symbols (e.g., **404**) at a plurality of symbol positions on a display device wherein the plurality of symbols correspond to an output amount for a play of an electronic game, presenting **1104** the output amount for the play of the electronic game, and determining **1106** that the plurality of symbols includes a plurality of consolidating symbols (e.g., **408A-C**).

[0117] Method **1100** also includes causing display **1108** of the plurality of consolidating symbols consolidating into a consolidated symbol (e.g., **502**) at a symbol position of the plurality of symbol positions previously occupied by a consolidating symbol wherein the consolidated symbol is associated with a number of future plays of the electronic game, causing **1110** any vacant symbol positions of the plurality of symbol positions to be filled with new symbols (e.g., **504-508**) wherein the new symbols, the consolidated symbol, and any remaining symbols of the plurality of symbols define a second plurality of symbols, determining **1112** a second output amount for the play of the electronic game based upon the second plurality of symbols, and presenting **1114** the second output amount for the play of the electronic game.

[0118] Method **1100** further includes, for each symbol position of the plurality of symbol positions other than the symbol position including the consolidated symbol, causing display **1116** of a third plurality of symbols for one of the number of future plays, the third plurality of symbols and the consolidated symbol defining a fourth plurality of symbols, determining **1118** a third output amount based upon the fourth plurality of symbols, and presenting **1120** the third output amount for the one of the number of future plays.

[0119] In some embodiments, method **1100** includes determining that the plurality of symbols includes a plurality of consolidating symbols by determining that the plurality of symbols includes a plurality of Wild symbols wherein the plurality of Wild symbols comprises the plurality of consolidating symbols. In some embodiments, method **1100** includes determining a respective output amount each time a consolidated symbol is consolidated into the consolidated symbol

wherein the plurality of consolidating symbols are consolidated into the consolidated symbol one-by-one. In some embodiments, method **1100** includes consolidating the plurality of consolidating symbols into the consolidated symbol one-by-one in an order such that each successive respective output amount is greater than each previous respective output amount.

[0120] While the disclosure has been described with respect to the figures, it will be appreciated that many modifications and changes may be made by those skilled in the art without departing from the spirit of the disclosure. Any variation and derivation from the above description and figures are included in the scope of the present disclosure as defined by the claims.

Claims

1. An electronic gaming device comprising: a display device; a memory device storing instructions; and a processor in communication with the display device and the memory device, wherein the instructions, when executed by the processor, cause the processor to: cause display of a plurality of symbols at a plurality of symbol positions on the display device, wherein the plurality of symbols correspond to an output amount for a play of an electronic game; present the output amount for the play of the electronic game; determine that the plurality of symbols includes a plurality of consolidating symbols; cause display of the plurality of consolidating symbols consolidating into a consolidated symbol at a symbol position of the plurality of symbol positions previously occupied by a consolidating symbol, wherein the consolidated symbol is associated at least with a number of future plays of the electronic game; cause display of a counter icon in association with the consolidated symbol, wherein the display of the counter icon includes display of a number of spins associated with the number of future plays; cause the consolidated symbol to persist at the symbol position across the number of future plays; and cause the counter icon to decrement the number of spins in response to each future play of the number of future plays.
2. The electronic gaming device of claim 1, wherein the display of the counter icon further includes display of a multiplier icon, and wherein the number associated with the number of spins is further associated with a multiplier value associated with the multiplier icon.
3. The electronic gaming device of claim 2, wherein the decrement of the number of spins includes decrementing a value of the multiplier value.
4. The electronic gaming device of claim 2, wherein the instructions further cause the processor to: for a winning symbol combination that includes a subset of the plurality of symbols and the consolidated symbol, apply the multiplier value to each output amount presented by the electronic gaming device that corresponds to the winning combination.
5. The electronic gaming device of claim 2, wherein the instructions further cause the processor to, for each future play of the number of future plays, decrement a value of the multiplier value in direct correspondence with the decrement of the number of spins.
6. The electronic gaming device of claim 1, wherein the memory device further stores one or more lookup tables including symbol data corresponding to the plurality of symbols.
7. The electronic gaming device of claim 6, wherein the symbol data includes data corresponding to at least one candidate consolidated symbol type, and wherein the instructions further cause the processor to: execute a consolidated symbol lookup in the one or more lookup tables; select the at least one candidate consolidated symbol type as the consolidated symbol into which the plurality of consolidating symbols are to be consolidated; and cause display of the selected at least one candidate consolidated symbol type as the consolidated symbol.
8. The electronic gaming device of claim 6, wherein the symbol data includes data corresponding to a plurality of replacement symbols, and wherein the instructions further cause the processor to: execute a replacement symbol lookup in the one or more lookup tables; select one or more replacement symbols of the plurality of replacement symbols for populating one or more vacated symbol positions vacated by the plurality of consolidating symbols as part of the plurality of

consolidating symbols consolidating into the consolidated symbol, the one or more vacated symbol positions including symbol positions other than the symbol position of the consolidated symbol; and cause display of respective replacement symbols of the one or more replacement symbols in respective vacated positions of the one or more vacated symbol positions.

9. A non-transitory computer-readable storage medium with instructions stored thereon that, in response to execution by a processor, cause the processor to: cause display of a plurality of symbols at a plurality of symbol positions on a display device, wherein the plurality of symbols correspond to an output amount for a play of an electronic game; transmit the output amount for the play of the electronic game; determine that the plurality of symbols includes a plurality of consolidating symbols; cause display of the plurality of consolidating symbols consolidating into a consolidated symbol at a symbol position of the plurality of symbol positions previously occupied by a consolidating symbol, wherein the consolidated symbol is associated at least with a number of future plays of the electronic game; cause display of a counter icon in association with the consolidated symbol, wherein the display of the counter icon includes display of a number of spins associated with the number of future plays; cause the consolidated symbol to persist at the symbol position across the number of future plays; and cause the counter icon to decrement the number of spins in response to each future play of the number of future plays.

10. The non-transitory computer-readable storage medium of claim 9, wherein the display of the counter icon further includes display of a multiplier icon, and wherein the number associated with the number of spins is further associated with a multiplier value associated with the multiplier icon.

11. The non-transitory computer-readable storage medium of claim 10, wherein the decrement of the number of spins includes decrementing a value of the multiplier value.

12. The non-transitory computer-readable storage medium of claim 10, wherein the instructions, in response to execution, further cause the processor to: for a winning symbol combination that includes a subset of the plurality of symbols and the consolidated symbol, apply the multiplier value to each output amount presented by the electronic gaming device that corresponds to the winning combination.

13. The non-transitory computer-readable storage medium of claim 10, wherein the instructions, in response to execution, further cause the processor to, for each future play of the number of future plays, decrement a value of the multiplier value in direct correspondence with the decrement of the number of spins.

14. The non-transitory computer-readable storage medium of claim 9, further storing one or more lookup tables including symbol data corresponding to the plurality of symbols.

15. The non-transitory computer-readable storage medium of claim 14, wherein the symbol data includes data corresponding to at least one candidate consolidated symbol type, and wherein the instructions, in response to execution, further cause the processor to: execute a consolidated symbol lookup in the one or more lookup tables; select the at least one candidate consolidated symbol type as the consolidated symbol into which the plurality of consolidating symbols are to be consolidated; and cause display of the selected at least one candidate consolidated symbol type as the consolidated symbol.

16. The non-transitory computer-readable storage medium of claim 14, wherein the symbol data includes data corresponding to a plurality of replacement symbols, and wherein the instructions, in response to execution, further cause the processor to: execute a replacement symbol lookup in the one or more lookup tables; select one or more replacement symbols of the plurality of replacement symbols for populating one or more vacated symbol positions vacated by the plurality of consolidating symbols as part of the plurality of consolidating symbols consolidating into the consolidated symbol, the one or more vacated symbol positions including symbol positions other than the symbol position of the consolidated symbol; and cause display of respective replacement symbols of the one or more replacement symbols in respective vacated positions of the one or more vacated symbol positions.

- 17.** A computer-implemented method of electronic gaming implemented by a processor in communication with a memory, the computer-implemented method comprising: causing display of a plurality of symbols at a plurality of symbol positions on a display device, wherein the plurality of symbols correspond to an output amount for a play of an electronic game; presenting the output amount for the play of the electronic game; determining that the plurality of symbols includes a plurality of consolidating symbols; causing display of the plurality of consolidating symbols consolidating into a consolidated symbol at a symbol position of the plurality of symbol positions previously occupied by a consolidating symbol, wherein the consolidated symbol is associated at least with a number of future plays of the electronic game; causing display of a counter icon in association with the consolidated symbol, wherein the display of the counter icon includes display of a number of spins associated with the number of future plays; causing the consolidated symbol to persist at the symbol position across the number of future plays; and causing the counter icon to decrement the number of spins in response to each future play of the number of future plays.
- 18.** The computer-implemented method of claim 17, further comprising displaying a multiplier icon in association with the counter icon, wherein the number associated with the number of spins is further associated with a multiplier value associated with the multiplier icon.
- 19.** The computer-implemented method of claim 18, further comprising decrementing a value of the multiplier value as part of the decrementing of the number of spins.
- 20.** The computer-implemented method of claim 17, further comprising storing, in the memory, one or more lookup tables including symbol data corresponding to the plurality of symbols, wherein the one or more lookup tables include at least one of a consolidated symbol lookup table and a replacement symbol lookup table.
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